The New
Drive Family

System overview • April 2003

sinamics

SIEMENS

Trends in drive technology

Two major trends are evident in drive technology:

- The variety of products is permanently increasing in machine building and plant construction, with the result that drive solutions must have a high degree of **flexibility** and **scaleability.**
- Customers in all industrial sectors are increasingly requesting solutions which are **simple to use** and **tailored** to individual requirements.

For drive technology this means that:

- Only those components and functions should be used which are required for the specific task.
  The drives are engineered according to different tasks, allowing cost-optimized and specific solutions, irrespective of whether single-axis or multi-axis applications are involved, whether simple speed control or highly dynamic servo control.
- The engineering costs for configuration and start-up must be kept as low as possible.
  Uniform tools for selection, configuration and start-up allow fast, simple and cost-effective engineering.
- Innovative approaches must be applied to all aspects. Distributed, intelligent drive technology permits a new approach to machine building and plant construction.

The new SINAMICS® drive family from Siemens provides a platform with which these requirements can be met.

SINAMICS is not only of great benefit to

- machine building and plant construction, it also sets new standards for
- the process industry and
- building technology.

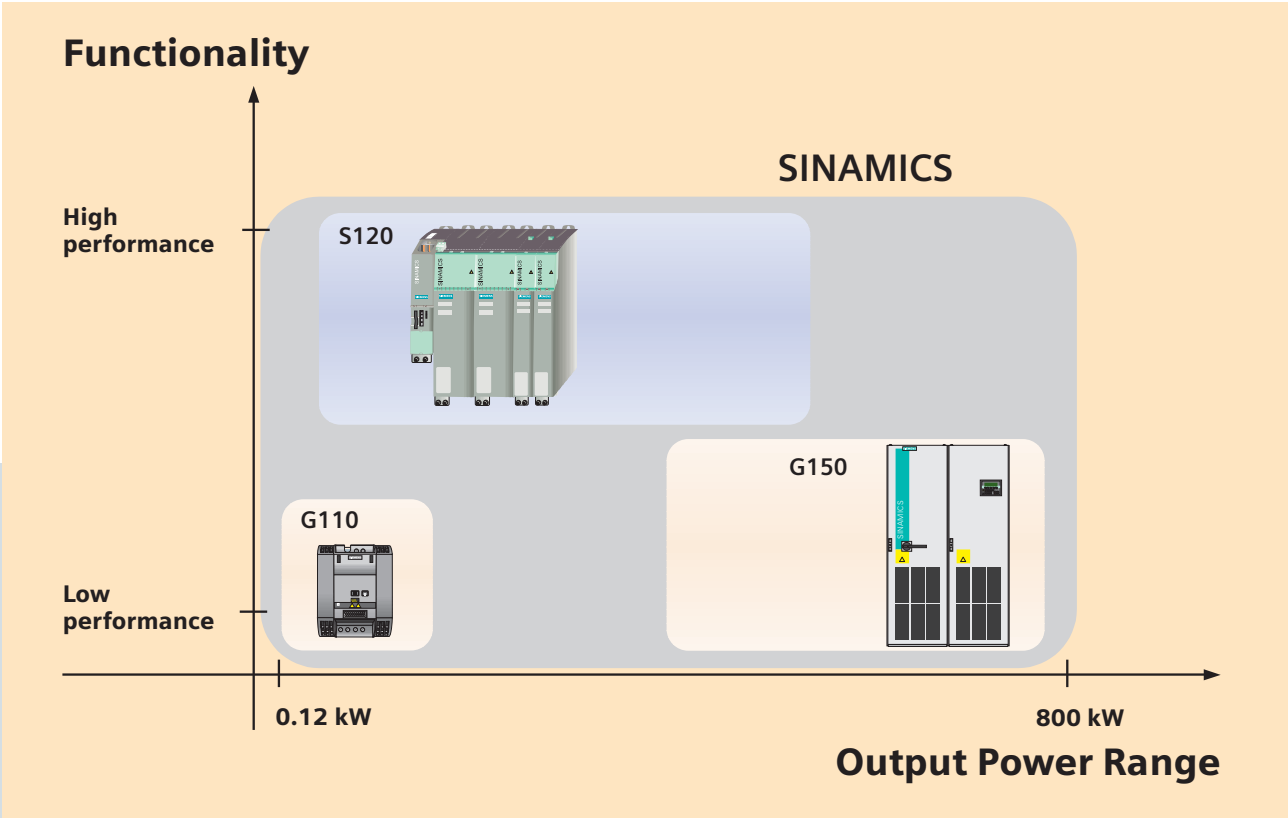
The SINAMICS drive family

The SINAMICS family comprises three members tailored to the respective application fields:

- ***SINAMICS G110*** – the versatile drive for small power ranges outputs
- ***SINAMICS G150*** – the powerful specialist for pumps, fans and turbo-compressors
- ***SINAMICS S120*** – the universal, modular drive system for demanding tasks

SINAMICS is characterized by the following system features:

- Uniform functionality based on a common platform strategy
- Uniform engineering
- High degree of flexibility and combination
- Wide output power range
- Designed for worldwide use
- SINAMICS Safety Integrated
- Increased economy and effectiveness
- Flexible interfacing facilities to host controllers
- Totally Integrated Automation.



The SINAMICS system

Uniform functionality based on a common platform strategy

The individual SINAMICS members are based on a common platform strategy: with SINAMICS, identical functions are based on the same software and hardware components. Functions such as drive control, operation, diagnostics or communication with host controllers are implemented uniformly for all members, facilitating handling of the drive engineering. There are synergies within the SINAMICS family in each individual drive member which enables training requirements to be reduced and support, maintenance and the stocking of spare parts to be simplified.

Uniform engineering

The SINAMICS family features high engineering uniformity and usability. The SIZER configuration tool provides user support for the selection and dimensioning of the drive components. Start-up and diagnostics are supported by STARTER.

This results in synergies and efficiency throughout the range with respect to configuration, parameterization and maintenance.

High degree of flexibility and combination

The coverage of a wide range of applications coupled with the high degree of commonality for functions and engineering make the SINAMICS family a universal platform for drive tasks. Components from various members of the family can, depending on the task, be flexibly combined into a complete drive solution.

Wide output power range

The SINAMICS family covers a wide output power range. A finely graded range of components allows optimally tailored drive solutions.

Designed for worldwide use

The SINAMICS drive family with its various members can be connected to all types of power supplies to be found worldwide.

It additionally satisfies international standards and regulations. Post-certification when exporting can therefore be eliminated.

A powerful, worldwide network of Siemens offices and partners is available for support.

Uniformity with SINAMICS

Uniformity	SINAMICS		
	G110	G150	S120
Configuration			
Parameterization			
Operation			
Diagnostics			
Communication			

SINAMCIS Safety Integrated

The SINAMICS family offers for the first time autonomous drive safety functions. Available in addition to "safe standstill" is for the first time "safe brake control" directly in the power module. The integration of these safety functions allows practicable safety concepts and simultaneously simplified installation. All safety functions are certified to international standards (IEC 61508, EN 954-1).

Improved economy and effectiveness

Drive solutions based on SINAMICS offer the prerequisites for improved cost effectiveness and competitiveness in machine building and plant construction as well as for end users:

- Solutions optimally tailored to the task as result of various drive members
- Reduced project execution times/order processing as a result of increased convenience and uniformity for configuring and start-up
- Maintenance-friendly, highly available machine and plant as result of uniform, powerful diagnostics mechanisms as well as reduced stocking of spare parts because of an optimized component range.

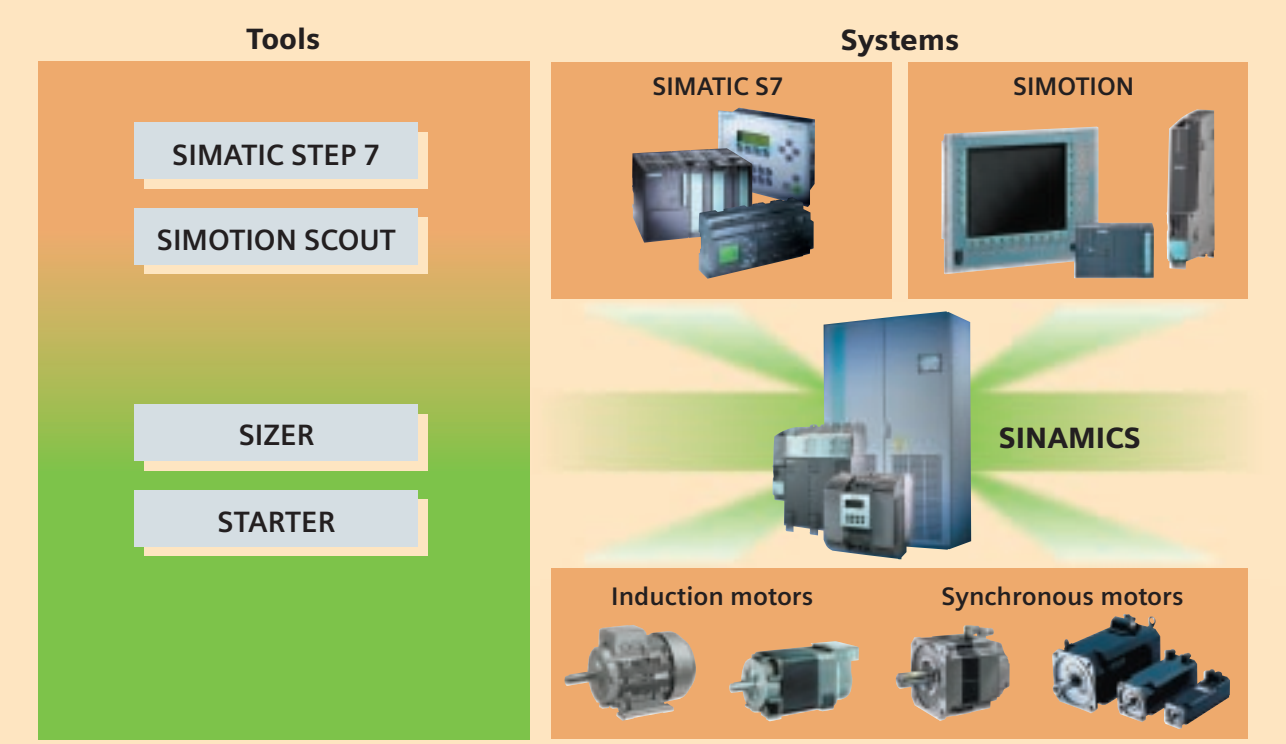
Flexible interfacing facilities to host controllers

SINAMICS offers several possibilities for interfacing to host controllers, including:

- Interfacing via digital or analog interfaces
- Interfacing to SIMATIC® or SIMOTION® via PROFIBUS DP

Totally Integrated Automation

SINAMICS is a component of the Siemens "Totally Integrated Automation". The uniformity of SINAMICS for configuring, data storage and communication to the automation level guarantees effortless solutions with SIMATIC and SIMOTION.



SINAMICS is a component of Totally Integrated Automation

Uniform engineering

SIZER configuration software

A particular feature of SINAMICS is that the user is supported by powerful and uniform tools during all phases of the project.

This begins with the dimensioning of the drive components. The tool covers the configuration of a complete drive system, and permits handling of single drives through to the complex multi-axis drives. The workflow wizard guides the user step-by-step through the individual configuration phases.

SIZER provides support for

- calculating the load,
- dimensioning of the motor and power components,
- configuration of further system components and

generates

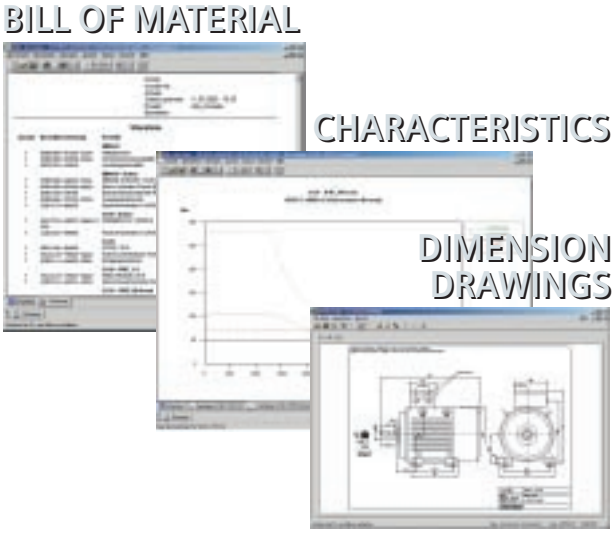
- configuration results (characteristics, technical specifications, location diagram and dimension drawings)
- and a bill of material with the associated ordering data.

Dimensioning of appropriate components in prompted steps, shown here with an example of the motor wizard

A guided tour helps first users to become acquainted with SIZER. The online help provides support during configuration with comprehensive physical and technical background information.

Non-plausible combinations are therefore eliminated, and incorrect orders avoided.

Electronic ordering of components is supported by an integral software/data exchange interface, e.g. to SAP systems.

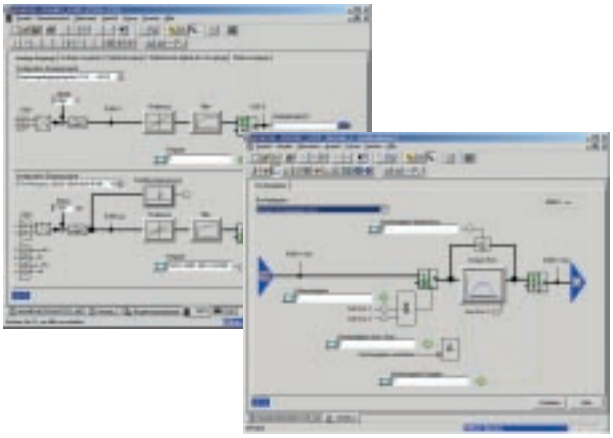


Result of configuration operation, e.g. bill of material, characteristics and dimension drawings

STARTER start-up software

All drives of the SINAMICS family may be commissioned using STARTER. The tool permits simple, menu-based and graphically-supported configuration and start-up of the drive components.

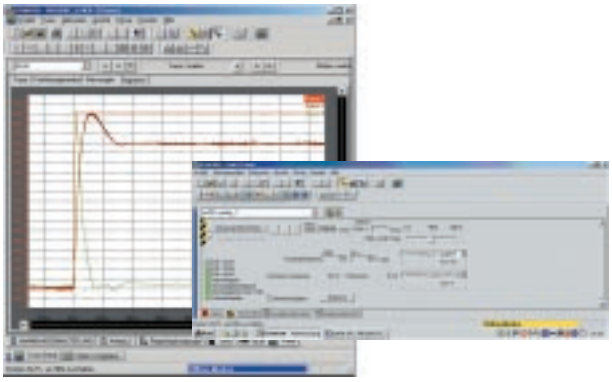
The import function for importing data from the electronic rating plates of the drive components significantly reduces parameterization overhead. This also reduces time requirements, and avoids incorrect inputs.



Graphic screens for drive configuration and start-up

The integral test functions can be used to check functions and to optimize parameters. The timing of speed, setpoint and actual-value curves can be recorded.

Graphics enable clear diagnostics and rapid user orientation.



The logging function records setpoint and actual-value curves

Both tools can be run as separate Windows applications. Interfacing to the drives is via the serial interface or PROFIBUS DP.

In conjunction with the SIMOTION motion control system, STARTER can also be integrated into the SCOUT engineering system. The same applies to operation of the drives together with the SIMATIC industrial automation system. In this case, embedding in the STEP 7 engineering software leads to complete integration of drive engineering into the PLC world.

The combination of SINAMICS with SIMOTION and SIMATIC results in fully integrated automation solutions from one source which can be configured, parameterized and started-up using one integral engineering software. If necessary simple diagnostics and troubleshooting can be carried out locally or via Teleservice.

The SIZER and STARTER tools are available in English and German.

SINAMICS G110 – the versatile drive for small power outputs

Overview

SINAMICS G110 is an ac drive suitable for a whole range of industrial variable speed drive applications. The very compact SINAMICS G110 provides voltage/frequency control (V/f) and is the ideal ac drive solution in the lower output power and performance ranges of the SINAMICS product family.

The drive is available in three footprint sizes, covers an output power range from 0.12 to 3.0 kW, and is suitable for connection to single-phase 200 to 240 V networks.

Applications

The SINAMICS G110 is particularly suitable:

- As a drive in industrial and commercial applications
- In various sectors, e.g. foods, textiles, packaging
- For conveyor applications
- For pumps and fans applications
- For factory gate, garage door and barrier openers
- As a drive for scrolling advertising billboards.

Highlights

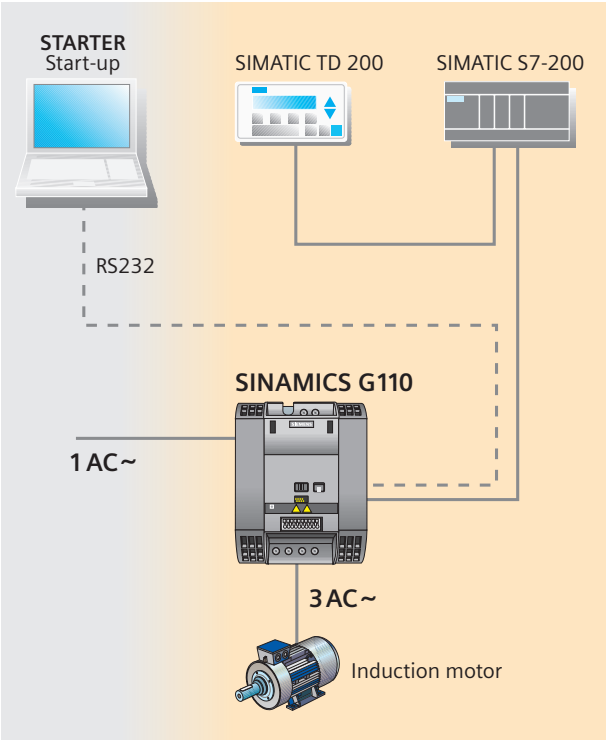
- Flexibility resulting from comprehensive parameterization facilities and various interfaces (analog and USS versions)
- Simple installation, parameterization and start-up
- Powerful diagnostics facilities with optional operator panel
- Fast standard system start-up using the optional operator panel and parameter cloning function
- Low-noise motor operation as a result of the high pulse frequency
- Low mechanical wear with
  – frequency band skipping to avoid resonance
  – programmable ramp-up/down times
  – ramp smoothing and
  – connection of drive to rotating motor (flying restart)
- Increase in plant availability as a result of automatic restarting following power failure or operating fault
- Fast current limitation for fault-free operation resulting from sudden load surges
- Versions with integral EMC filter for industrial and residential networks
- DIP switches for fast adaptation to 50 or 60 Hz power supplies
- DIP switches for simple bus termination for the USS version (RS485)

Design

SINAMICS G110 units have a compact design and are supplied ready for connection. All units have state-of-the-art IGBT technology in the power unit as well as digital microprocessor technology. SINAMICS G110 units are easy and fast to install and hook-up.

SINAMICS G110 is available with an analog input or with an RS485 communication interface (USS). One version with a cooling-fin free heat sink is particularly suitable for installation in shallow control cabinets. The digital inputs are freely programmable and permit flexible adaptation to a wide variety of applications.

The SINAMICS G110 is programmed either via a PC using STARTER or using an optional operator panel. For a standard system start-up of several drives with the same parameters, the entered settings can be saved in to the operator panel and copied to each further drive.



Configuration example with SINAMICS G110 (USS version together with SIMATIC S7-200, connection between PC and drive using optional PC connection kit)

Technical specification	
Electrical data	
Supply voltages; output ranges	200 to 240 V 1 AC, ± 10%; 0.12 to 3.0 kW
Supply systems	IT, TN, TT
Line frequency	50/60 Hz
Motor output frequency	0 to 650 Hz
Control methods	V/f control, linear (M~n) V/f control, quadratic (M~n²) V/f control, programmable
Fixed frequencies	3, programmable
Skipped frequency bands	1, programmable
Digital inputs	3 programmable digital inputs 24 V DC
Analog input	Analog version: 1 analog input for setpoint, scaleable from 0 to 10 V
Digital output	1 digital output 24 V DC
Communication interface	USS version: RS485 serial interface for operation with USS protocol
Functions	
Software functions	Automatic restart following interruptions in operation due to a power failure, Smooth connection of drive to rotating motor, Programmable ramp-up/down times, Ramp smoothing
Protection functions	Undervoltage, overvoltage, earth fault, short-circuit, stall prevention, thermal motor protection I²t, drive overtemperature, motor overtemperature
Suitable motors	Induction motors
Mechanical data	
Degree of protection	IP20
Type of cooling	≤ 0.75 kW: Convection cooling, or version with flat heat sink > 0.75 kW: Internal air cooling (integrated fan)
Standards	
Conformity	CE, UL, cUL, c-tick

SINAMICS G150 – the powerful specialist for pumps, fans and turbo-compressors

Overview

SINAMICS G150 are ready to connect and simple to use drive cabinet units. They have a clear, modular design specially tailored for drives with a quadratic load torque.

These drive applications do not usually require a high overload reserve and dynamic response, and can therefore be operated using a relatively simple control without speed feedback.

SINAMICS G150 takes exactly these aspects into account, providing a cost-effective drive solution which can be matched to specific customer requirements using a wide range of options.

SINAMICS G150 covers an output power range from 75 to 800 kW.

Applications

In industrial applications it is advantageous to use variable-speed drives as opposed to fixed-speed drives wherever solid, liquid or gaseous media have to be moved, conveyed, pumped, compressed or sucked.

The main reasons for this are the

- optimization of process control,
- increase in yield, and especially
- significant energy savings, particularly in the partial load range.

SINAMICS G150 drive converter cabinet units are particularly suitable for these variable-speed drive applications with a quadratic load torque $M \sim n^2$ such as

- pumps,
- fans and
- turbo-compressors.

Highlights

- Particularly quiet and compact drives in standard control cabinets using state-of-the-art IGBT power semiconductors and innovative cooling solution
- Problem-free interfacing into automation solutions using various analog and digital interfaces
- Simple start-up and parameterization using interactive menus on the cabinet's operator panel
- Maximum servicing friendliness with easy accessibility to all modules
- Increase in plant availability through easy, fast replacement of individual modules and power components

Design

SINAMICS G150 drive cabinet units are available with cabinet widths from 400 mm upwards in intervals of 200 mm, in two design versions, and in various degrees of protection up to IP54.

The user-friendly operator panel contains an LCD-Display incorporating graphics capability for menu prompting and plain text displays.

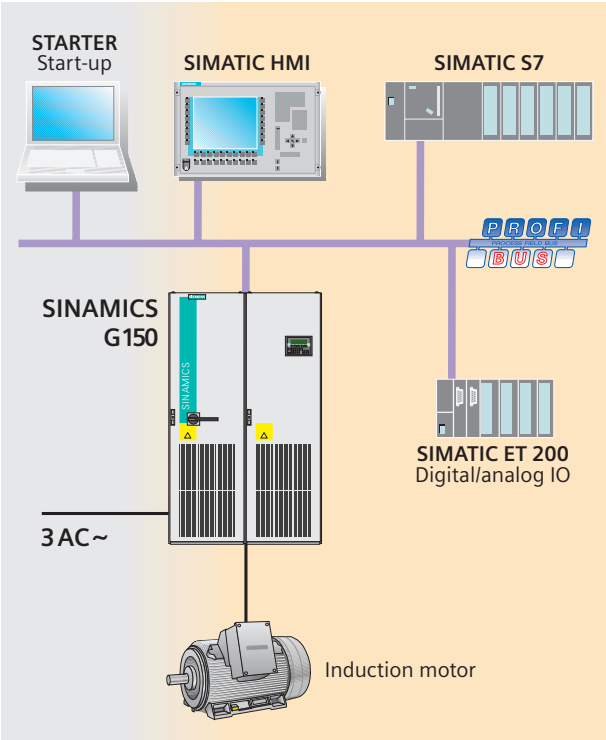
Soft function keys reduce the number of keys required and facilitate operator prompting. Bargraphs indicate the most important process variables at a glance. Comprehensive auxiliary and diagnostics functions describe faults in detail including their cause and tracing possibles.

Version A

offers sufficient space for all the options available.

Version C

is a space-saving version and is intended for cases where the power supply components are accommodated in a central low-voltage distribution panel and therefore do not need to be installed in the drive control cabinet.



Configuration example with SINAMICS G150 and SIMATIC S7

Technical specification	
Electrical data	
Supply voltages; output ranges	380 to 480 V 3 AC, ± 10 % (–15 % < 1 min); 110 to 560 kW 660 to 690 V 3 AC, ± 10 % (–15 % < 1 min); 75 to 800 kW
Supply systems	IT, TN, TT
Line frequency	50/60 Hz
Motor output frequency	0 to 83 Hz
Control method	Sensorless vector control
Fixed frequency	3 fixed frequencies plus 1 basic frequency, programmable
Skipped frequency bands	4, programmable
Digital inputs	8
Bidirectional digital inputs/outputs	4
Analog inputs	2, differential
Analog outputs	2
Relay outputs	2, changeover contacts
Speed setpoint	10 V, 0 to 10 V, 4 to 20 mA, ± 20 mA
Communication interface	PROFIBUS DP
Functions	
Software functions	Automatic restart following interruptions in operation as a result of a power failure, Smooth connection of converter to rotating motor, Programmable ramp-up/down times, Ramp smoothing
Protection functions	Undervoltage, overvoltage, earth fault, short-circuit, stall prevention, thermal motor protection I^2t, drive overtemperature, motor overtemperature
Suitable motors	Induction motors
Mechanical data	
Degree of protection	IP20, optionally IP21, IP23, IP54
Type of cooling	Internal air-cooling (forced air cooling with integral fan)
Noise level	≤ 72 dB (A) with 50 Hz line frequency ≤ 75 dB (A) with 60 Hz line frequency
Cabinet system	Rittal TS 8
Standards	
Conformity	CE

SINAMICS S120 – the universal, modular drive system for demanding tasks

Overview

SINAMICS S120 is the modular drive system within the SINAMICS drive family for demanding applications, e.g. for dynamic and highly dynamic positioning and motion control applications in machine building and plant construction.

The drive system is suitable for both single-axis and multi-axis applications. Since the SINAMICS S120 even supports simple V/f control in addition to vector and servo control, it is possible to universally solve all drive tasks in a single machine: speed control, positioning tasks and – in conjunction with SIMOTION – motion control tasks up to and including high-accuracy axis synchronization.

With outputs from 1.6 up to 107 kW, the SINAMICS S120 covers the output power range found in dynamic and highly dynamic applications.

SINAMICS S120 has powerful communication ports for interfacing to host controllers.

Together with SIMOTION or SIMATIC, it is possible to design fully integrated and powerful machine automation solutions from one supplier.

Applications

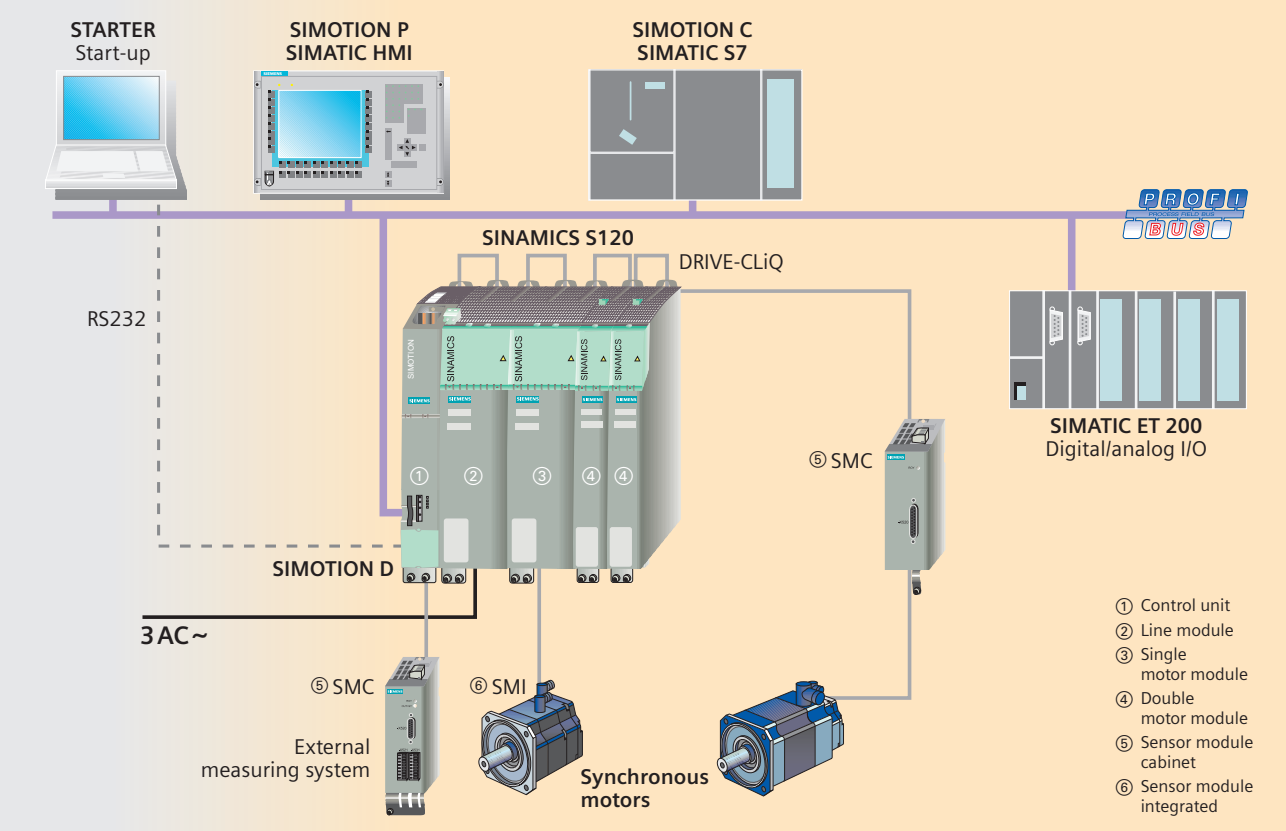
The SINAMICS S120 features make it fully suitable as a drive system for all types of production machines such as:

- Packaging machines
- Machines for foods processing
- Plastics processing machines
- Textile machines
- Presses, punching machines
- Printing and paper machines
- Machines for the wood processing, glass and ceramics industries
- Automatic assembly and testing units
- Handling machines
- Hoisting

Highlights

The SINAMICS S120 components were developed for installation in cabinets. They feature:

- Easy handling
- Simple installation and wiring
- Practical connection system
- Uniform design
- Gap-free mounting
- Internal air cooling
- SINAMICS Safety Integrated.



Configuration example with SINAMICS S120

Design

SINAMICS S120 has an innovative, flexible and modular system architecture for multi-axis applications.

Control units

- Provide computing performance as well as interfaces to the I/O and host controls
- Available with various performance levels

Line modules

- Supply the current from the power supply network
- Available for outputs from 5 to 120 kW

Motor modules

- Provide the connected motors with power
- Available for motor currents from 3 to 200 A or output ratings from 1.6 to 107 kW

Electronic options/sensor modules

- Used to interface supplementary components such as external position measuring systems to the drive

DRIVE-CLiQ

- Powerful system interface for data exchange between individual SINAMICS S120 modules
- All SINAMICS S120 components, including motors and encoders, are connected together via the common DRIVE-CLiQ serial interface. The uniform design of the cable and connector systems reduces the range of parts and storage costs
- Interface modules are available for non-Siemens motors or retrofit applications for conversion of conventional encoder signals to DRIVE-CLiQ

A control unit, a line module and one or more motor modules are combined to form a drive line-up. All modules are linked together via DRIVE-CLiQ. The DC-links of the line module and motor module(s) are additionally connected by busbars.

During the planning phase, SIZER supports the selection of the required drive components.

Parameterization is carried out using STARTER.

Digital rating plate in all components

All SINAMICS S120 components have a digital rating plate containing all relevant technical specifications. For motors, for example, these include the parameters of the electrical equivalent circuit diagram and the characteristics of the integral motor encoder. These data are automatically read by the control unit via DRIVE-CLiQ, and need not be re-entered during start-up or following replacement.

In addition to the technical specifications, the digital rating plate also contains logistic data such as the manufacturer ID, the order no. and the unique worldwide ID number. Since these values are available electronically on site and also by remote diagnostics, unique identification of all components used in a machine is possible at all times simplifying servicing.

SINAMICS S120 – the universal, modular drive system for demanding tasks

Technical specification

Electrical data	
Supply voltages; output ranges	380 to 480 V 3 AC, ± 10%; 1.6 – 107 kW
Supply systems	IT, TN, TT
Line frequency	50/60 Hz
Control methods	V/f control Vector control with and without encoder Servo control Dynamic servo control (DSC) for highly dynamic applications with external motion control
Digital inputs/outputs	Yes, scaleable number
Analog inputs/outputs	Yes, scaleable number
Communication interface	Digital inputs/outputs, 24 V DC Analog inputs/outputs PROFIBUS DP CANopen Serial (RS232)
Functions	
Drive control	V/f control Simplified positioning (available soon) Motion Control in connection with SIMOTION D Vector control with and without encoder Servo control

Functions (continued)	
Technological functions	Free interconnection of setpoints and actual values In-process measurement Tracing kinetic cushioning
Safety functions (Safety Integrated)	Safe Standstill Safe brake control Further safety functions in connection with SIMOTION D (available soon)
Limitations	Torque limitation Current limitation Speed limitation Direction limitation Min./max. speed setpoint limitation
Protection functions (extract)	Thermal monitoring of motor and power modules Overcurrent, overvoltage, undervoltage, stall protection, over speed, zero speed
Setpoint preparation	Speed reversal 4 skip bands Simple ramp-function generator without smoothing, with special fast stop ramp Extended ramp-function generator with smoothing and setting function Speed setpoint filter
Suitable motors	Induction motors Synchronous motors
Supported encoders/ direct measuring systems	Resolver EnDat absolute-value encoder SSI absolute-value encoder Incremental encoder, sin/cos 1 V_{pp} Incremental encoder, TTL signal, RS422 Incremental encoder, HTL
Mechanical data	
Degree of protection	IP20
Type of cooling	Internal air cooling
Standards	
Conformity	CE, cULus

The members of the SINAMICS family

	SINAMICS G110	SINAMICS G150	SINAMICS S120
	The versatile drive for small power outputs	**The powerful specialist for pumps, fans and turbo-compressors**	**The universal, modular drive system for demanding drive tasks**
Main applications			
	• Machines and plant for industrial and commercial applications	• Machines and plant for industrial applications (water/waste water, power plants, oil and gas, petro-chemicals, basic chemical industry, paper, cement, mining, steel)	• Machines and plant for industrial applications (packaging, plastics, textiles, printing, wood, glass, ceramics, presses, paper, hoisting equipment, semiconductors, automatic assembly and testing units, handling)
Application examples			
	• Pumps • Fans • Auxiliary drives • Conveyor belts • Billboards • Gate/door openers	• Pumps • Fans • Turbo-compressors	• Synchronous speed control • Positioning • Motion control • Axis interpolation
Highlights			
	• Compact • Flexible adaptation to different applications • Simple, fast start-up • Ready-to-use • Clearly arranged terminals • Optimum interaction with SIMATIC & LOGO!	• Space-saving • Low noise • Simple, fast start-up • Cabinet unit ready to connect • Optimum interaction with SIMATIC	• For universal use • Flexible, modular • Scaleable power, function, performance, number of axes • Simple, fast start-up, auto-configuration • Innovative system architecture • Wide range of motors • Optimum interaction with SIMOTION and SIMATIC • SINAMICS Safety Integrated
Degree of protection			
	IP20	IP20, optionally IP21, IP23, IP54	IP20
Line voltages			
200 – 240 V 1AC	0.12 – 3 kW	–	
380 – 480 V 3AC	–	110 – 560 kW	1.6 – 107 kW
660 – 690 V 3AC	–	75 – 800 kW	–
Control method			
V/f control	Yes	–	Yes
Sensorless vector control	–	Yes	Yes
Vector control	–	–	Yes
Servo control	–	–	Yes

The members of the SINAMICS family

	SINAMICS G110	SINAMICS G150	SINAMICS S120
Motors			
Induction motors	Yes	Yes	Yes
Synchronous motors	–	–	Yes
Functions			
Variable speed	Yes	Yes	Yes
Torque control	–	–	Yes
Simplified positioning	–	–	Yes
Synchronism	–	–	Yes
Axis synchronization	–	–	Yes
Safety Integrated	–	Yes	Yes
Communication interfaces			
Digital	Yes	Yes	Yes
Analog	Yes	Yes	Yes
Serial	Yes	Yes	Yes
PROFIBUS DP	–	Yes	Yes
CANopen	–	–	Yes
Start-up, parameterization, diagnostics			
Using operator panel	Yes	Yes	Yes
Using PC (STARTER)	Yes	Yes	Yes
Further information			
	www.siemens.com/ sinamics-g110	www.siemens.com/ sinamics-g150	www.siemens.com/ sinamics-s120 www.siemens.com/ simotion

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